Farès Zaghouane

Machine Learning Engineer

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SUMMARY

My work sits between representation learning and dynamical systems. At Apple I formalized hardware behavior as reachable-state specifications; at Nokia Bell Labs I built video-prediction models for cloud gaming, where predictions drift once the semantic anchor becomes sparse. That failure mode led me to the question I work on now: how the representation space and the predictor of a world model should be designed together so that planning becomes more stable, adaptable, and computationally efficient.

RESEARCH INTERESTS

Keywords: World models · JEPA predictors · latent dynamical systems · state-space models · Koopman-style spectral structure · energy-based models · latent geometry · representation learning · planning and control.

In joint-embedding predictive architectures, the encoder learns what to represent while the predictor models how representations evolve; that predictor is typically an unconstrained network. I tested whether explicit dynamical structure helps on a system where the prior is exactly correct after a change of basis: with the representation held fixed it did not, and part of the penalty comes from the coordinates in which the prior is imposed rather than from the transition family itself. What I want to build next is a diagnostic framework that reveals when structure pays: which modes persist, how much stability margin remains before rollouts diverge, and which latent directions actions can reach.

EDUCATION

École Polytechnique Fédérale de Lausanne (EPFL) Lausanne, Switzerland

MSc Computer Science, Machine Learning specialization *Sept 2023 – 2026*

Relevant coursework: Artificial Neural Networks & Reinforcement Learning, Computer Vision, Machine Learning, Natural Language Processing, ML in Finance, Cryptography & Security.

CPE Lyon Lyon, France

Diplôme d'Ingénieur, Computer Science — Software Engineering & Big Data track *Sept 2021 – 2026*

Relevant coursework: Software architecture, Design patterns, Databases, Data mining, Embedded systems, System administration, Quantum computing.

CPE Lyon Lyon, France

Licence (BSc), Computer Science & Electronics *Sept 2020 – Sept 2021*

Relevant coursework: Linear algebra, Algorithms, Signal processing, Digital & analog electronics, Discrete & continuous mathematics, Quantum mechanics.

EXPERIENCE

Nokia Bell Labs Stuttgart, Germany

Research Intern (MSc Thesis) *Aug 2025 – Feb 2026*

- An under-specified research problem: first decide what to predict and how to measure whether it helps. The target became dense focus maps predicted from cloud-gaming video, to allocate bandwidth and quality under a strict per-frame latency budget.
- No dataset existed for the task. With the team, built a metric that makes training possible without hand-annotated ground truth, and a data engine that first fine-tunes a VLM to annotate the data.
- Started from a JEPA-style model propagating the latent state to predict the next focus maps; dropped it under the latency, compute and dataset constraints, and derived a teacher–student pipeline combining Grounding DINO semantic anchoring, RAFT optical-flow propagation, and a lightweight correction network.
- Also tried state-space models and Mamba-Vision encoders; designed the pseudo-supervision setup, tracked temporal drift and built the dense-map evaluation protocol. The retained hybrid met the real-time latency budget, recovered most of the semantic-only quality at a fraction of the cost, and outperformed motion-only and saliency-only baselines.

Apple Munich, Germany

Design Verification Research Intern *Oct 2022 – Jul 2023*

- Modeled IP behavior using finite-state machines and event-based abstractions, allowing long-running processes to be reasoned about without expanding every low-level timing step.
- Applied model checking to behaviors that simulation cannot cover exhaustively, and built Python tooling that generates Verilog/SystemVerilog assertions from high-level specs, validated on industrial designs; FSM-level specs feed directly into the model-checker.
- The event-based abstraction sets the level at which reachable states are specified, which changed which properties were practical to verify.

PROJECTS

Structured Latent Dynamics *(in progress)*

Factorial study of structured dynamical priors in the latent predictor of a JEPA world model: Koopman-style 2×2 rotation-scaling blocks against MLP, transformer and selective-SSM predictors on coupled oscillators, whose true generator belongs to the structured family after a change of basis. Arms are paired within backbone. In the first cycle, structure does not buy rollout stability, on two backbones independently. Least squares recovers the planted operator, so the transition family is not what fails: the prior is a statement about a basis, not about coordinates, and an oracle-basis arm traces part of the penalty to that basis without reversing the sign.

Constrained Trace EBM *(in progress)*

Energy-based scoring of finite-state-machine execution traces against temporal specifications, with an exact symbolic checker as ground truth and decision thresholds fixed before training. In the first cycle, detection on unseen machines clears its floor (bootstrap lower bound 0.772 against 0.75), but a classifier that never sees the properties scores higher (0.905); the next cycle targets the tier that needs the specification.

RL from Scratch

From-scratch PyTorch RL curriculum: 25 algorithms from tabular Q-learning to PPO/SAC, Dreamer, MuZero, and an action-conditioned JEPA world model inspired by V-JEPA 2-AC, each taught in a course-style notebook then refactored onto a shared runtime. HalfCheetah-v5: SAC/TD3 ≈10k, DDPG ≈11k near 1M steps.

Machine Learning from Scratch

From-scratch ML curriculum: 52 course-style notebooks and a 123-module scikit-learn-style library. Each chapter takes one model apart rather than only re-implementing it: every equation term, every hyperparameter, and why one design supersedes the last. Runs from a custom autograd engine through Transformers, SSMs, GNNs and generative models to optimal transport and Riemannian optimization.

V-JEPA2 ROI Semantic Extraction

Research prototype that organizes V-JEPA-style predictive video representations around semantic regions rather than uniform spatial patches.

GPT-2 from Scratch

PyTorch GPT-2-style decoder-only Transformer built against the original papers: tokenization, embeddings, causal multi-head attention, transformer blocks, pretraining, generation, fine-tuning.

VLM Data Engine

Video data pipeline for multimodal world-model experiments: shard processing, FPS conversion, scene detection, clip extraction, annotation, and quality checks.

SKILLS

ML & representation learning: PyTorch, TensorFlow, Neural networks, Self-supervised learning, JEPA / world models, Energy-based models

Vision & multimodal systems: Computer vision, Video understanding, Grounding DINO, RAFT / optical flow, Focus-map prediction, Multimodal data pipelines

Dynamics, control & RL: Reinforcement learning, State-space models, Mamba, Koopman-style spectral constraints, Dynamical systems

Mathematical & scientific foundations: Linear algebra, Applied statistics, Signal processing, Optimization, Spectral methods, Functional analysis

Software & formal methods: Python, NumPy, Formal verification, Model checking, Finite-state machines, Verilog / SystemVerilog assertions